

Solvan Enterprise Architecture

The authoritative architecture description of the Solvan autonomous production engineering platform: system context, runtime and logical structure, data authority, integration seams, security, isolation, deployment, reliability, operations, quality governance, and the decision record — written to the standard an enterprise architecture review board expects, and verifiable against the repository it describes. Solvan was built for the All Things Agentic hackathon and is engineered to outlive it as Ruhu Inc.'s production reliability platform for its wider product line.

How to read this document. Every named component, table count, test name, and boundary is checkable in the repository at the pinned revision. Status vocabulary follows `docs/documentation-policy.md`: *implemented* means a production path exists; *verified* means named deterministic checks passed; *target* means a reviewed normative contract not necessarily in the current release. Nothing labelled target is offered as evidence of implementation.

Companion document. The illustrated competition reference (`Solvan_Competition_Architecture.html`) presents the same system through the lens of the hackathon's assessment material. This document is the standalone engineering reference; where the two overlap, both derive from the same specifications and tests.

Python 3.12

Google ADK 2

Gemini on Vertex AI

Agent Runtime

PostgreSQL 16 · Cloud SQL

Cloud Run

Pub/Sub · Cloud Scheduler

Terraform

React · TypeScript

OpenTelemetry · Cloud Trace

Document control

Version	1.0
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Owner	Solvan engineering
Status	verified against repository audit
Audience	engineering, security review, platform operations, enterprise assessment

Normative sources

- `specs/01-22` — behavior contracts (17,218 lines)
- `specs/artifacts/schema.sql` — authoritative DDL
- `AGENTS.md` — engineering invariants
- `ARCHITECTURE.md` — concise runtime map
- `docs/documentation-policy.md` — authority and status

953

test functions behind a ten-stage merge gate with PR-001.050 requirement-to-test traceability enforced in CI.

Scope

the Solvan platform: control plane, agents, action path, data, channels, deployment

Out of scope

customer estate internals; Google-managed service internals; the demo fixture's business logic

Print setup

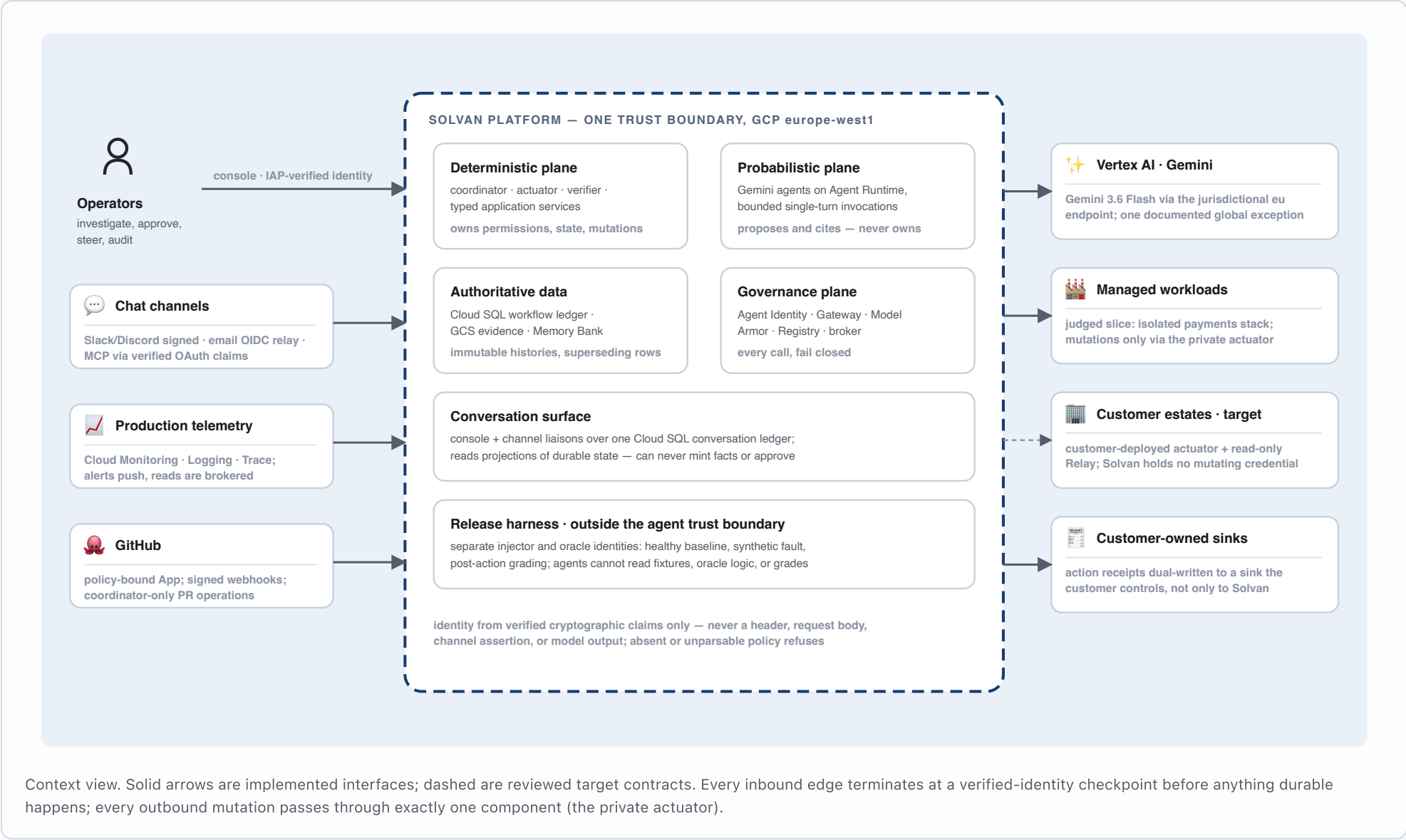
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Change rule

update the smallest governing specification with every behavior change

01 System Context And Scope

What Solvan is: a governed fleet of institutional agents that detects production degradation from events, investigates live evidence, executes bounded policy-authorized recovery, verifies the outcome independently, and owns a multi-day permanent-repair case. What surrounds it, and on what terms:



Interface	Direction	Protocol and authentication	Status
Monitoring alerts	in	Pub/Sub push with verified push identity → token-fenced inbox	implemented
Scheduler ticks	in	Cloud Scheduler → authenticated tick endpoints; the only driver of work	implemented
Operator console	in/out	authenticated HTTP; console is a projection surface, never an authorization boundary	implemented
Chat channels	in/out	Slack/Discord native signatures · email audience-bound OIDC relay · MCP verified OAuth claims (ask , catch_up , resolve_ref)	implemented · locally verified (spec 14)
Model inference	out	Vertex AI Gemini, region-pinned eu endpoint; optional deep workspace uses a documented global exception	implemented
Production mutation	out	private Action Actuator only: reservation → dry-run → actuate → receipt	implemented (judged slice)
GitHub operations	in/out	private regional provider service; signed webhooks; coordinator-only PR operations; no generic git push	implemented
Customer estate evidence	in	Solvant Relay: outbound poll for one signed job, closed-catalog read, local redaction before egress	target (spec 22)

02 Architecture Principles And Invariants

These are not aspirations: each principle is enforced by code, schema, or a required test, and changing one requires changing the governing specification. They bind every section that follows.

P1 • Two planes, one authority

Gemini may investigate, rank hypotheses, plan, summarize, and draft. Typed application services own permissions, state transitions, risk, action budgets, approval validity, target reservations, idempotency, and mutations. Model output can become a proposal — never authority.

P2 • Events start work, never prompts

Incidents begin from events. Chat can explain and request; only external events create lifecycle. Nothing in the platform wakes itself: Cloud Scheduler and Pub/Sub are the only drivers.

P3 • Cloud SQL is the only workflow authority

Agent Runtime executions, Sessions, Memory Bank, traces, model output, and connector responses are never workflow authority. Every decision reads typed rows; every prompt is rebuilt from them.

P4 • No agent invokes another agent

Agents return typed plans and results; the coordinator creates every durable `agent_run` and performs all dispatch. Isolation is structural, not configured, and cannot regress by editing a prompt.

P5 • Every mutation crosses the same gates

Deterministic Execution Agent → exact digest-bound authorization → target-level reservation → TOCTOU recheck → dry-run and effect comparison → receipt → independent verification. No second road exists.

P6 • Nothing verifies itself

No agent approves its own action or selects a weaker verification profile. Verification shares no identity, provider process, conversation, or mutable artifact with the producer it adjudicates. Proposer ≠ approver is enforced in code and SQL.

P7 • Identity from cryptographic claims only

Principal and scope are never taken from a header, request body, channel assertion, or model tool argument. A read-audience grant is never proof of write or steer authority. No configuration value may downgrade an identity, policy, or isolation control; absent or unparsable policy refuses.

P8 • Everything external is untrusted data

Logs, traces, repositories, tickets, tool responses, model output, and Memory Bank content are data, never instructions. Model Armor verdicts persist on evidence and gate memory promotion.

P9 • Histories are immutable

Incidents, approvals, receipts, verification results, and promoted memories are never rewritten; superseding records are added. Structured decisions and evidence references are recorded — never secrets, raw credentials, unredacted PII, or model chain-of-thought.

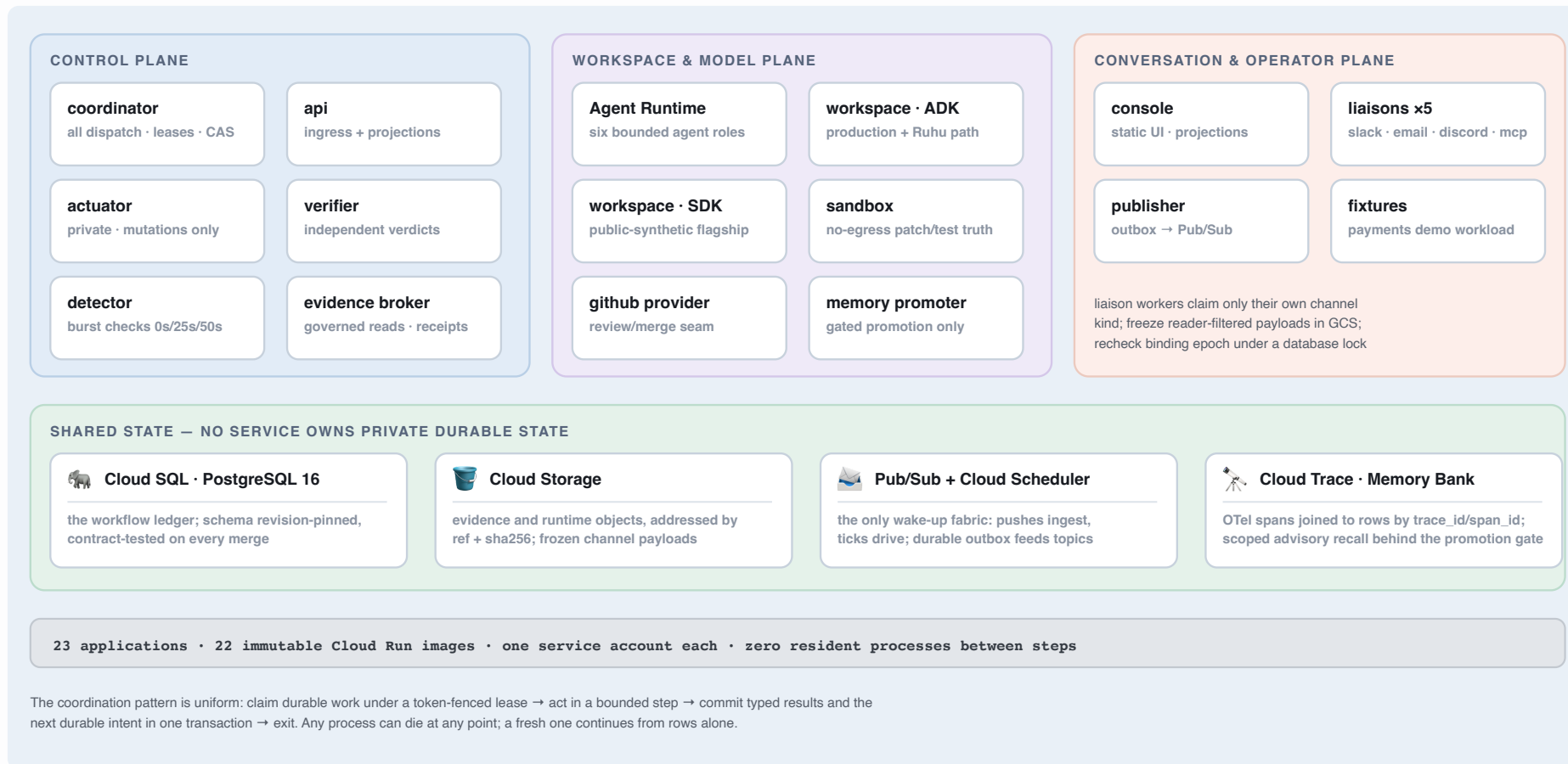
P10 • Sovereignty before retrieval, again before prompts

Tenant, project, environment, purpose, classification, and region filters are applied before retrieval and re-applied before prompt construction. Preview platform features carry tested degradation paths and are never the sole control on a consequential action.

Enforcement, not prose. The repository prefers deterministic enforcement in code, tests, hooks, or CI over adding prose. Where this document states a rule, the corresponding constraint lives in `specs/artifacts/schema.sql`, a typed service, or a named test cited in Sections 09–11.

03 Runtime View: Services And Planes

Twenty-three deployable applications, each a bounded, stateless Cloud Run service with its own service account and least-privilege IAM — no shared identities. Nothing is resident between steps: the platform's unit of execution is a bounded run resumed from rows.



Runtime view. Group membership is enforced by IAM, not convention: read-plane identities cannot reach mutation endpoints (`SEC-IAM-READONLY-001` , `SEC-ACTUATOR-IAM-001`), and the actuator is network-private.

04 Logical View: Packages And Dependency Rules

Package responsibilities

Package	Responsibility
domain	identifiers, states, transitions, risk, budgets, verification rules
application	incident/case use cases, orchestration, durable investigation plans, policy, approvals, memory promotion
persistence	Cloud SQL repositories, transactions, inbox/outbox, leases, reservations
agents	catalogued ADK institutional agents and bounded coordinator-dispatched roles
tools	immutable catalog revisions, capability profiles, typed read/compute/mutate contracts, connector wrappers
platform	Registry, Runtime, Sessions, Memory Bank, Identity, Gateway, Armor, OTel adapters
connectors	GCP telemetry/deployment, policy-bound GitHub App, PostgreSQL and synthetic-check adapters
api / console	authenticated ingress and projections / operator UI — never an authorization boundary
benchmarks	injectors, oracles, evaluators, immutable evidence receipts
infra	Terraform, IAM, network, regional placement, deployment preflight

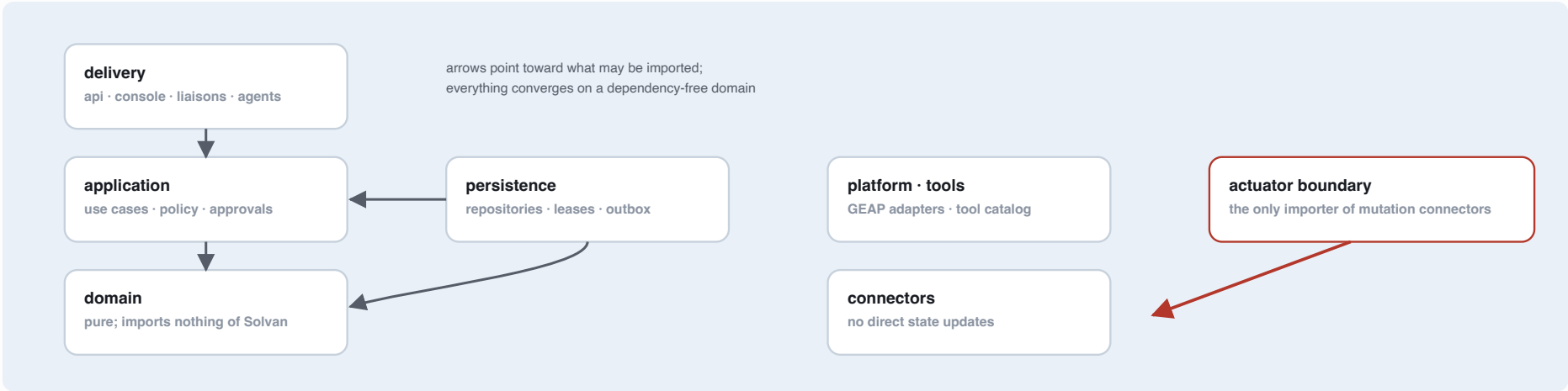
Dependency rules (architecture-tested)

- domain imports no Solvan package.
- persistence depends only on domain and database libraries.
- Agents and delivery adapters call application ; application code never imports the console or HTTP routes.
- Connectors cannot update incident or case state directly.
- Only the private Action Actuator can import mutation connectors; the Execution Agent may invoke only that actuator, with an action ID.
- Verification does not import remediation planning or mutation connectors.
- Memory promotion consumes confirmed domain events, never raw prompt text.
- Solvent Relay and the Action Actuator may share audited utility libraries, but neither executable may import the other's connector, route, identity, or repository boundary.

These rules are checked by the architecture-boundary stage of scripts/check ; a violating import fails the merge, not a review comment.

Adoption profiles

Workload-specific topology, signals, operations, and verification journeys load as adoption profiles that never alter the domain or grant permissions implicitly. Ruhu's GCP profile (spec 11) is the first: observe-only by default, with separately gated approval-bound and bounded-autonomous phases.



Dependency direction. The red edge is deliberately unique: exactly one boundary in the system may import mutation connectors, which is what makes "no second road to production" a compile-time property rather than a review habit.

05 Data Architecture And Authority Model

Release schema · 61 tables

- Incidents, cases, plans, steps, runs, tool calls, evidence, hypotheses, actions, approvals, reservations, receipts, transitions, wakeups, inbox/outbox, memory, workspaces.
- Loaded from empty into clean PostgreSQL 16 in CI (`scripts/check-contracts`); constraint oracles assert negative cases; append-only audit triggers.

Operability schema · 54 tables

- Governed Tool Catalog and capability profiles; Operational Guidance revisions with registered completion predicates; trigger policies; skills interchange records.
- Status: **implemented and locally verified** (specs 16–18); guidance selection, tool-catalog wiring, and quarantined skills import run in real coordinator paths. Cloud release qualification pending.

Conversation schema · 34 tables

- One canonical Liaison ledger across console and channels: read/steer grants, per-part access modes (an empty access set denies), durable delivery, compaction.
- Identity from verified claims only; a read grant never proves write or steer authority.
- Status: **implemented and locally verified** (spec 14); cloud release qualification pending.

Store	Holds	Authority	Integrity mechanism
Cloud SQL workflow ledger	every incident, case, run, action, approval, receipt, transition, wakeup	authoritative — the only tier any decision reads	CAS + workflow versions; token-fenced leases; partial unique indexes (one active attempt); append-only histories with superseding records
GCS evidence objects	telemetry extracts, artifacts, frozen channel payloads	citable evidence; never instructions	ref + sha256 addressing; classification, residency, freshness, retrieval actor, and armor verdict recorded per item
Memory Bank	promoted, scoped operational facts	advisory recall only	promotion requires type-specific confirmation rules plus a recorded armor verdict (<code>armor_verdict_ref</code> NOT NULL); revocable; quarantine for poisoned candidates
ADK Sessions	interaction context for a bounded invocation	none — never resumed as truth	rebuilt from authoritative rows for every fresh fenced attempt
Conversation ledger	liaison threads, compaction summaries	durable projection; reads state, never mints it	compaction may summarize talk but cannot rewrite plans, evidence, approvals, receipts, or workflow state (spec 03 §14.1)

Further target schemas — alert triage (28 tables, implemented), outcome quality (17), production graph (16), connection scope (4), relay (18), SaaS scale (28) — bring the platform's schema surface to 260 tables.

Idempotency — layered and schema-enforced

- Alert ingress unique on (`source`, `source_event_id`); duplicates return the original durable event.
- One active attempt per logical step (partial unique index); unique `invocation_id` and `provider_request_id`.
- Approvals bind an exact digest with expiry; execution re-checks state, version, and policy (TOCTOU) and releases without mutating on drift.
- Receipts unique per dispatch; undo plans derived from observed pre-state; action budgets and cooldowns bound repetition.

Data protection and sovereignty

- Tenant/project/environment scope columns on every row; classification, purpose, and residency filters applied before retrieval and again before prompt construction.
- Authoritative data is region-pinned to `europa-west1` ; a placement violation is a release-blocking policy failure.
- Tenant material under CMEK; stored long-lived credentials are read-only scoped Secret Manager references with recorded posture (target, spec 13).
- No secrets, raw credentials, unredacted PII, or model chain-of-thought in any log, row, or trace.

06 Integration View: Channels, Connectors, And Seams

Seam	Contract	Why it cannot escalate	Status
Conversational channels Slack · Discord · email · MCP · console	One Cloud SQL Liaison ledger; public Slack/Discord ingress verifies native signatures; email accepts only an audience-bound relay identity; MCP derives binding from verified OAuth claims and exposes exactly <code>ask</code> , <code>catch_up</code> , <code>resolve_ref</code> .	No external channel can approve or mutate production; channel workers claim only their own channel kind; payloads frozen reader-filtered in GCS; binding epoch rechecked under a database lock before delivery.	implemented · locally verified (spec 14)
Evidence tools	Broker-fronted, allow-listed, approved query signatures; every request writes one <code>tool_calls</code> row (cache hits included); bounded, citation-bearing receipts.	Read-only IAM; classification and residency filters precede execution; immutable call budgets; argument fingerprints make repeats return the prior receipt.	implemented
Production mutation	Private Action Actuator loads stored, approved actions by ID; reservation → TOCTOU recheck → dry-run and effect comparison → actuate → receipt dual-written to a customer-owned sink.	Agents never see credentials or payloads; the actuator reloads and revalidates all authority itself; receipts unique per dispatch.	implemented (judged slice)
GitHub release provider	Private regional Cloud Run boundary: signed webhook ingress, CI-published branch check, coordinator-only PR operations, independent review/check/head gates, receipt-backed Cloud SQL projection.	Not a model-facing tool; never performs a generic <code>git push</code> ; never exposes credentials to the console.	implemented
Solvent Relay	Separate customer-resident read-only product: polls outward for one coordinator-authored signed job, performs one closed-catalog read under customer-signed local policy, redacts locally, returns one idempotent evidence projection.	Separate executable, image, identity, audience, registry, API, repository, and deployment boundary from the actuator; contains no mutation capability and cannot gain one through configuration.	target (spec 22)
Workspace providers	Two implementations behind one internal <code>WorkspaceAgent</code> contract: Google ADK + regional Agent Runtime (production/Ruhu), and the pinned official <code>google-antigravity==0.1.10</code> SDK self-hosted on private regional Cloud Run (public-synthetic flagship). Exact patch/test truth comes only from the separate no-egress Cloud Run Sandbox in <code>europa-west1</code> .	The Alpha SDK provider receives only inputs classified PUBLIC and independently attested <code>synthetic=true</code> ; no repository secrets, telemetry, customer data, credentials, or GCS/SQL authority. Both providers return proposals only.	implemented
Agent Skills interchange	The open Agent Skills format is adopted as a quarantined interchange profile: packages are hardened, scanned, held as inert records; a human compiles them into governed guidance via evaluation and independent approval.	Scripts never execute; <code>allowed-tools</code> grants nothing; activation pins content and never fetches origins.	implemented · locally verified (spec 18)

The uniform rule at every seam. Discovery is never authorization; a connection is never authority; a transport is resolved by the coordinator from an exact registered binding, never selected by a model.

07 Security Architecture

Identity and access

- Distinct SPIFFE principal per agent; every Cloud Run service has its own service account with least-privilege IAM, proven by negative tests (SEC-IAM-READONLY-001 , SEC-ACTUATOR-IAM-001).
- Human identity established from verified cryptographic claims; approvals additionally require the APPROVER role, proposer ≠ approver, and an unexpired digest-bound grant.
- Grants for reads and proposals on a person's behalf are short-lived and audience-bound, minted from verified identity — never inferred from headers, bodies, or channel assertions.

Model and content safety

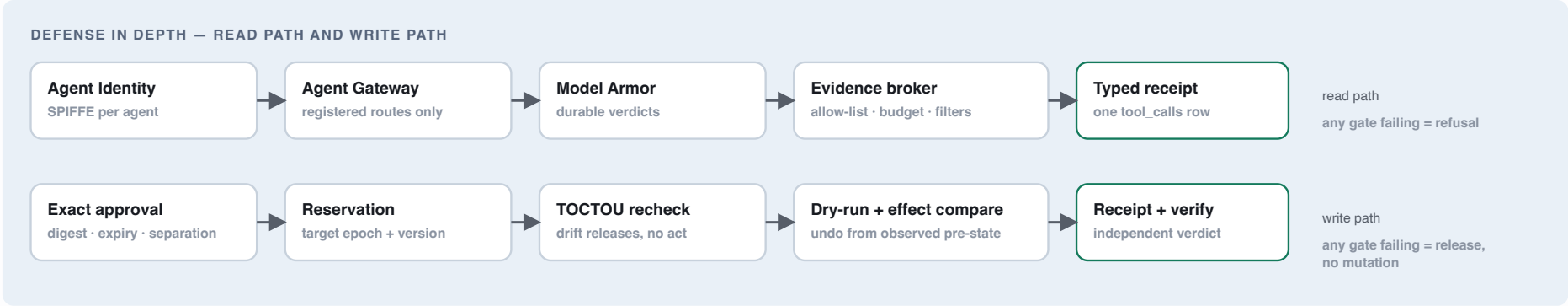
- All external content — logs, traces, repositories, tickets, tool output, model output, memory — is data, never instructions; agent prompts state it and the broker enforces it.
- Model Armor screens supported paths inline; verdicts persist durably on evidence and gate memory promotion; a poisoned candidate is quarantined, not retried into memory.
- Operator-facing factual claims instantiate from an enumerated, digest-pinned template registry; a failed predicate is withheld, never softened into prose.

Secrets and credentials

- Solvan's target posture: never hold a credential capable of mutating a customer estate; mutation capability lives only in a customer-deployed actuator under customer policy, kill switch, and rate budget (spec 13).
- Every stored credential records its posture; federated short-lived credentials preferred; stored long-lived keys are read-only scoped Secret Manager references under per-tenant CMEK, surfaced in the console.
- Development and key-file hosts are never production-eligible by default.

Fail-closed configuration

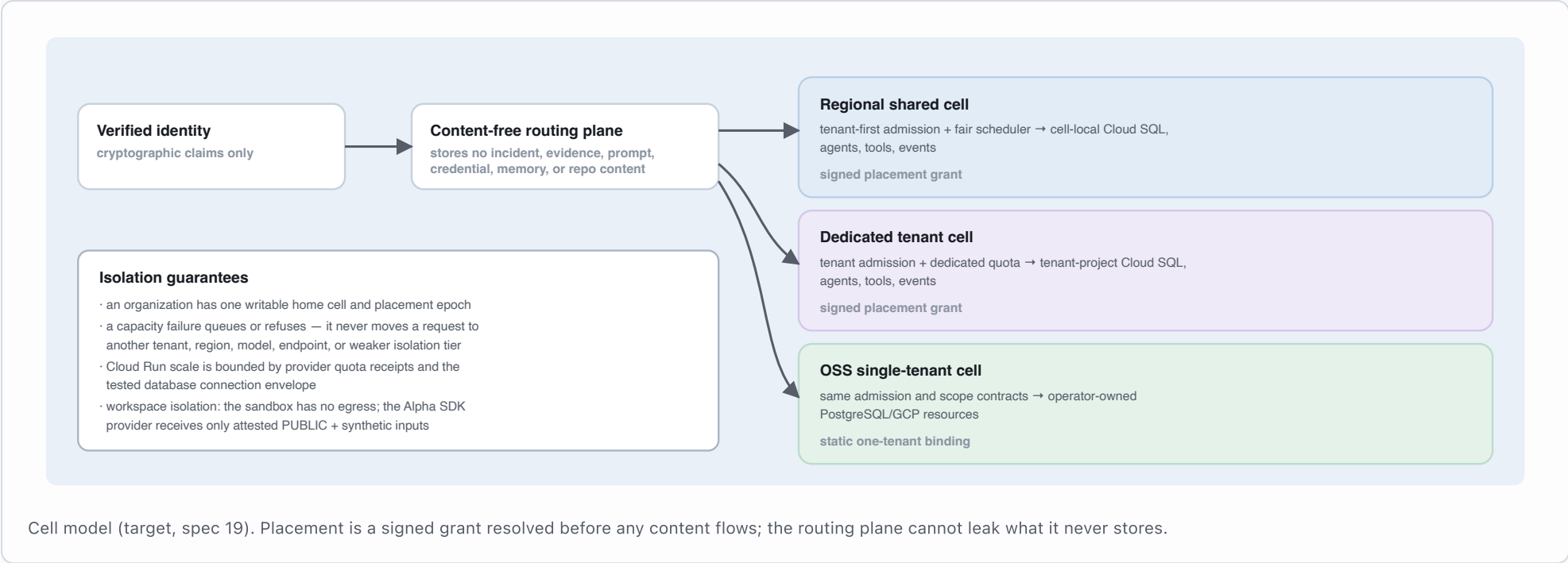
- No configuration value may downgrade an identity, policy, or isolation control to a permissive mode; absent or unparsable policy refuses.
- The Agent Gateway is registry-governed and fail-closed with no generic mutation route; bypass denial is a deployed test (SEC-GATEWAY-BYPASS-001) re-proven by deployment preflight (gateway_bypass_denied).
- Preview platform features carry tested degradation paths and are never the sole control protecting a consequential action.



Both paths are chains of independent gates with no shared discretion: the components that propose, approve, execute, and verify are different identities with different IAM, and the review-time threat model (stale attempts, duplicate delivery, cross-incident target races, restart recovery, memory poisoning, cross-tenant access) has a named test per row in Section 10.

08 Tenancy, Isolation, And The Cell Model

The competition deployment is one cell. The open-source and managed product uses the same runtime inside the placement and capacity boundary of specification 19 — a reviewed target contract, drawn here as such.



Isolation inside one cell today. The judged deployment already enforces the in-cell contracts: tenant/project/environment scope on every row, per-agent identity, verification separated from producers, the no-egress sandbox, and the public-synthetic boundary for the Alpha SDK provider.

09 Deployment, Environments, And Release Path

Local worktree

- `scripts/bootstrap` installs locked dependencies; `scripts/start` runs an isolated stack: this worktree's PostgreSQL, API, console, logs, metrics, traces.
- Ports, databases, processes, and artifacts are scoped per worktree through `scripts/dev-env`; no production authority exists locally.
- Compose/state names carry the schema revision; prior-revision volumes are preserved during DDL expansion.

GCP dev

- Mutable engineering environment in `eu-west-1`; never release proof.
- Same Terraform stack (`infra/terraform/environments/gcp`) with its own project, backend prefix, identities, secrets, and data resources.

GCP staging

- The authoritative release, rehearsal, recording, and submission environment.
- Isolated project; region-pinned; deployment preflight validates agents, identities, gateway policies, armor templates, IAM matrix, and region policy before any release exists — a missing control blocks the deploy.

Release step	Mechanism	Evidence
Build	Cloud Build produces 22 immutable images from the pinned revision.	<code>cloudbuild.yaml</code> contract, checked in CI
Preflight	<code>tools/deploy_release.py</code> validates the governance plane before traffic: registry entries, per-agent identity, gateway routes and bypass denial, armor templates per region, IAM matrix, region policy.	stored preflight receipts; <code>gateway_bypass_denied</code>
Deploy	Cloud Run services + Agent Runtime resources updated as a unit; each service keeps its own identity.	service revisions pinned in the release record
Verify	Isolated baseline/fault/action/oracle release scenario against live GCP, including pool-recycle recovery and demo rehearsal from a clean environment.	immutable release receipts
Boundary rule	Local harness evidence never counts as a GCP release receipt; cloud proofs are captured separately at deploy time.	<code>AGENTS.md</code> canonical-commands contract

Region posture. All authoritative state and control-plane services are pinned to `eu-west-1`. Fast-fleet Gemini inference uses the documented `eu` multi-region via `aiplatform.eu.rep.googleapis.com`; the optional Gemini 3.1 Pro Preview deep workspace is the single documented `global` location exception, and it is not load-bearing for any consequential action.

10 Reliability And Continuity

Solvan's continuity model is structural: there is no long-lived process whose loss can lose work. Suspension is a database state; recovery is a fresh, fenced attempt rebuilt from rows; repeated effects are impossible by constraint rather than unlikely by care.

Failure	Deterministic recovery	Named proof
Worker dies holding an inbox claim	Claim expiry re-opens the work; the dead worker's token is fenced and can no longer complete it.	<code>test_expired_inbox_claim_is_recovered_and_old_token_is_fenced</code>
Process dies after durable create, before dispatch	The tick sweeps expired CREATED runs into failure + the declared fallback path.	<code>_recover_expired_created_runs</code> · reserve/expire logic
Process dies after dispatch, before the result	Any fresh process polls the durable operation name, validates, and accepts the result exactly once — no session state required.	<code>test_dispatched_runtime_operation_is_recovered_by_fresh_process_without_session_state</code>
Poison event crash-loops its handler	Bounded claim budget (5 inbox / 25 outbox) parks the event as quarantined: visible, audited, inert.	<code>test_crash_looping_inbox_event_is_quarantined_after_bounded_claims</code>
Transient provider fault on dispatch	Durable retry backoff (<code>retry_not_before</code> as a column) defers the single declared fallback across restarts and workers.	<code>test_declared_agent_fallback_is_attempted_once_after_dispatch_failure</code>
Contention at scale	Twenty-five simultaneous incidents are neither lost nor duplicated.	<code>test_twenty_five_simultaneous_incidents_are_not_lost_or_duplicated</code>
Silent dormancy of long work	Case liveness policy: every operational state must hold a next action and an active wakeup; stale progress escalates instead of rotting.	case machine YAML · <code>test_mitigated_incident_opens_one_case_and_due_wakeup_is_fenced</code>

Continuity of multi-week work

Reliability Cases (15 states) and `scheduled_wakeups` carry work across days with zero resident compute: an 18-hour approval pause or a 7-day case survives any deployment or crash because it is rows, not a parked invocation. Workspace continuity uses hibernated checkpoints rehydrated under exact provider revision and boot-proof validation.

What is deliberately not promised

No mid-invocation resume: a half-finished model invocation is never trustworthy state and is always discarded in favor of a fresh fenced attempt. Upstream exactly-once is never claimed without a separately qualified provider idempotency contract; Solvan's own exactly-once claims are per-dispatch receipts and schema constraints, and stop at its boundary.

11 Observability And Operations

Telemetry model

- OpenTelemetry across every service; spans exported to Cloud Trace and joined to the domain timeline via `trace_id / span_id` columns on runs — a judge or operator can walk from an incident row to the exact trace and back.
- Structured decision records with reason codes and budget states; stop reasons are recorded, chain-of-thought never is.
- Every tool call, cache hit included, is one `tool_calls` row with fingerprint and receipt.

Operator surfaces

- Console: incidents, cases, fleet, approvals (target, digest, environment, risk, expiry), evidence (source, freshness, classification, armor verdict), release evidence — control made visible.
- Validated versus inferred findings are visually and semantically distinct; an unresolvable citation renders inert and demotes the statement to inferred.
- Local parity: `scripts/observe [status|logs|traces]` inspects any worktree's isolated stack.

Operational guidance

- Runbooks and skills live as governed data with registered completion predicates (specs 17–18): guidance versions are approved, pinned, and evaluated — never free prose a model can drift. Implemented and locally verified.
- Autonomy is an earned, revocable receipt derived from outcome quality, not a configuration default (spec 20).

Alert operations

- Canonical event ingress normalizes Cloud Monitoring webhooks and detector bursts into idempotent durable events; alert triage (spec 21) runs on the same fenced-claim machinery as everything else.
- Nothing pages a human by default: escalation is a policy outcome with a durable record, and abandoned work escalates via the stale-progress policy.

Auditability as a feature. Because histories are append-only and superseded rather than rewritten, the answer to "who decided what, on which evidence, under which authority" is a query, not an investigation.

12 Quality And Release Governance

The one authoritative merge check

1. Python lint and format
2. Python types (mypy, 315 files)
3. Architecture boundaries, agent manifests, Ruhu profile, Cloud Build contract
4. Documentation and repository knowledge
5. Executable schema + transition contracts on clean PostgreSQL 16
6. Unit and harness tests with coverage (953 test functions)
7. Agent safety and trajectory evaluation
8. Console types
9. Console production build
10. Browser critical path

`scripts/check` is the single gate; a failing canonical check is never bypassed with an alternate command — the harness is fixed or the result recorded as blocked.

Traceability and change discipline

- PR-001..050 requirement-to-test traceability enforced in CI: a missing or skipped required test fails the merge.
- Verification map: unit (transitions, digests, risk, budgets, redaction) · contract (agents, tools, registry, gateway, schemas) · integration (transactions, leases, approvals, plan-before-dispatch, Memory Bank scopes) · end-to-end (six competition acceptance scenarios) · security (negative IAM, bypass, injection, PII, tenant, region) · release (preflight, live GCP evidence, recovery, rehearsal).
- Every behavior change updates the smallest governing specification; safety-sensitive changes are reviewed against the standing threat checklist (stale attempts, duplicate delivery, target races, restart recovery, memory poisoning, cross-tenant access).
- No tool, permission, connector, model, region, or platform feature is added without updating its threat model, registry contract, and acceptance tests.

```
scripts/bootstrap    # locked dependencies and Chromium
scripts/start        # isolated stack: PostgreSQL, API, console, logs, traces
scripts/check        # the full ten-stage gate, including restart-recovery,
                    # idempotency, TOCTOU, quarantine, and approval-digest tests
```

Anyone with the repository can reproduce the quality claim in one command; the release claim additionally requires the stored GCP receipts described in Section 09.

13 Architecture Decision Record

The ten decisions that shape everything else, with the consequence each one buys and the cost it accepts. Full rationale lives in the cited specifications.

ID	Decision	Why	Accepted cost	Where
AD-01	Split the system into a probabilistic plane that proposes and a deterministic plane that owns all authority.	Model output can be wrong, injected, or stale; making it structurally non-authoritative removes the whole class of "the model did it" failures.	More typed contracts and validation code than a prompt-centric design.	spec 02, 03
AD-02	Make Cloud SQL the only workflow authority; Sessions, Memory Bank, and traces are never truth.	One authoritative ledger makes recovery, audit, and idempotency provable with schema constraints.	Prompts must be rebuilt from rows; nothing convenient may be read back from a transcript.	spec 04
AD-03	Coordinator-only dispatch: no agent invokes another agent.	Orchestration lives where it can be tested; isolation cannot regress by editing a prompt.	Multi-agent flows require durable plan validation instead of free delegation.	spec 03
AD-04	No mid-invocation resume; the unit of recovery is the durable <code>agent_run</code> .	A half-finished invocation's in-memory state is untrustworthy; fresh fenced attempts plus idempotency constraints are strictly safer.	Re-doing bounded work after a crash instead of resuming it.	spec 03
AD-05	Approval is a cryptographic act over an exact digest, with expiry and proposer/approver separation.	A remembered "yes" cannot authorize drifted content; TOCTOU rechecks make approval and execution one auditable pair.	Any drift after approval requires a fresh proposal round-trip.	spec 05
AD-06	Independent verification with zero shared context with producers.	Connector success is not recovery; self-confirmation is the classic autonomous-agent failure.	Every mitigation pays for a second, independent observation pass.	spec 05
AD-07	Mutation capability lives in exactly one private actuator; target: deployed customer-side so Solvan holds no estate-mutating credential.	Concentrating mutation makes "no second road" enforceable by IAM and import rules; customer-side deployment makes the blast radius customer-governed.	An extra deployment artifact and policy surface per customer.	spec 13
AD-08	Pin all authoritative state and control services to <code>europa-west1</code> ; fail closed on placement.	Data sovereignty must be a property of the topology, not a promise in prose.	A documented, non-load-bearing exception path for one global model endpoint.	spec 07, 20
AD-09	Immutable histories with superseding records; no rewrite of incidents, approvals, receipts, verdicts, or promoted memories.	Audit is a query; compaction and summaries can never corrupt authority.	Storage growth and superseding-record complexity.	spec 04
AD-10	Adopt open interchange (Agent Skills) only through quarantine, human compilation, and independent approval.	Interoperability without inheriting arbitrary third-party instructions as authority.	Slower import path than direct skill execution.	spec 18

14 Implementation Status And Roadmap

The honest claim, in three levels. The release-gated slice runs under enterprise controls and is the only basis of any release claim. A second tier is implemented and locally verified in this same codebase, with cloud release qualification deliberately pending. The remainder is specified: reviewed normative contracts with acceptance tests, not roadmap prose. Solvan continues beyond the competition as Ruhu Inc.'s production reliability platform.

Release-gated — the demonstrated slice

- Event-driven incident lifecycle with durable coordination, recovery, quarantine, and multi-day cases.
- Bounded agent fleet on Agent Runtime with typed contracts and structural isolation.
- Action safety path: exact approvals, reservations, TOCTOU rechecks, dry-run, receipts, independent verification.
- Governance plane: per-agent SPIFFE identity, fail-closed Gateway with tested bypass denial, Model Armor verdicts, Registry-driven routing.
- Data authority model: 61-table release schema, evidence objects, gated Memory Bank.
- Region-pinned dual-project deployment with preflight-gated releases.
- Ten-stage merge gate with PR-001..050 traceability (953 tests).

Implemented — locally verified, qualification pending

- **Conversational surface** (spec 14): five channel liaisons on one ledger — Slack lifecycle, steer confirmation, subscriptions, deterministic compaction.
- **Governed Tool Catalog** (16) and **Operational Guidance** (17): hash-bound selection and computed completion in real coordinator dispatch.
- **Agent Skills interchange** (18): quarantined import; local implementation gate passed.
- **Alert triage** (21): 28-table schema, fenced-claim runtime, console surfaces.
- **Production environment model** (20): production graph and earned-autonomy receipts substantially implemented.
- None of this tier is part of any release claim until cloud qualification runs with valid authority.

Specified — target contracts

- **SaaS cells and isolation** (spec 19): content-free routing, placement grants, admission — implementation under way.
- **Solvent Relay** (spec 22): customer-resident read-only evidence collection; specification complete, implementation early.
- **Customer-deployed actuator** (spec 13): Solvan never holds an estate-mutating credential.
- **Multi-tenant sovereignty**: per-tenant CMEK across a tenant population (depends on 19).
- **Ruhu adoption profile** (spec 11): observe-only → approval-bound → bounded autonomy, receipt-gated phases.

Enterprise property	Mechanism	Status
Data sovereignty	Classification + residency filters before retrieval and before prompts; region-pinned storage; CMEK for tenant material.	implemented (judged scope) / target (multi-tenant)
Credential posture	Recorded posture per credential; federated short-lived preferred; read-only long-lived keys under CMEK, surfaced in console.	target (spec 13)
Auditability	Append-only transitions and audit events; structured decisions; no chain-of-thought anywhere.	implemented
Fail-closed configuration	No config value may downgrade identity, policy, or isolation; absent or unparsable policy refuses.	implemented
Preview-feature honesty	Launch stages recorded with retrieval dates; preview features carry tested degradation paths.	implemented

15 Glossary

Incident	The durable record of one detected degradation, driven by a 14-state machine from DETECTED to a verified terminal state.
Reliability Case	The multi-day, 15-state follow-up that owns permanent repair after mitigation; subject to the liveness policy.
agent_run	The durable unit of model work: one bounded attempt with pinned hashes, budget, deadline, and provider operation name — created before dispatch.
Bounded invocation	The complete authority ceiling for one agent attempt: scope, owner, allowed tools, deadline, budget, evidence refs, trace context.
Token-fenced lease	A claim on durable work that expires; a stale holder's token can no longer commit results.
CAS transition	Compare-and-set state change bound to expected state and workflow version; the only way workflow state moves.
TOCTOU recheck	Re-validation of state, versions, epoch, and policy at execution time; any drift releases the reservation without mutating.
Receipt	The unique-per-dispatch durable record of an actuation, dual-written to a customer-owned sink.
Evidence object	A hash-addressed GCS artifact with classification, residency, freshness, retrieval actor, and armor verdict.

Promotion gate	The rule set (type-specific confirmation + recorded armor verdict) between confirmed events and Memory Bank.
Liaison	A channel worker (Slack, email, Discord, MCP, console) reading reader-filtered projections of one conversation ledger.
Workspace	A logical Cloud SQL/GCS record executed through bounded stateless provider tasks; no provider process or filesystem is durable state.
Sandbox	The separate no-egress Cloud Run service that produces patch/test truth; the only source of "tests passed."
Actuator	The single private service allowed to import mutation connectors and touch production, by stored action ID only.
Verifier	The independent service computing recovery verdicts from fresh telemetry and synthetic probes; shares nothing with producers.
Cell	The placement and capacity boundary holding one tenant population's runtime and data (spec 19; target).
Placement grant	The signed decision binding an organization to its one writable home cell and epoch.
Adoption profile	A workload-specific contract (signals, topology, journeys) that never alters the domain or grants permissions implicitly.

Related reading: [README.md](#) (product entry point), [ARCHITECTURE.md](#) (concise map), [specs/](#) (normative contracts), [Solvan_Competition_Architecture.html](#) (illustrated competition reference).